

Multi-Objective Optimization

Salient Issues of Multi-Objective Evolutionary Algorithms

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Salient Issues of Multi-Objective Evolutionary Algorithms

Chapter Overview

1 Representation of Non-Dominated Solutions

- Scatter-plot matrix, value path, bar chart, star coordinate, visual methods

2 Performance Metrics for MOEAs

- Convergence metrics, diversity metrics, combined metrics

3 Test Problem Design

- ZDT problems, constrained test problems, problem generator

4 Comparative Studies of MOEAs

- Zitzler-Deb-Thiele, Deb-Agrawal-Pratap-Meyarivan studies

5 Objective vs Decision-Space Niching

6 Searching for Preferred Solutions

7 Exploiting MOEAs for Other Problems

8 Scaling Issues

9 Convergence Issues

10 Controlled Elitism

11 Multi-Objective Scheduling

Challenge: Visualizing > 2 Objectives

Representing Non-Dominated Solutions

Why This Matters:

- Most MOEA studies focus on 2 objectives (easier visualization)
- Real-world problems often have 3+ objectives
- Need methods to understand trade-offs in higher dimensions

Common Representation Methods:

Method	Best For	Key Feature
Scatter-Plot	3–5 objectives	Pairwise plots
Matrix		
Value Path	3+ objectives	Cross-lines show trade-offs
Bar Chart	Comparative analysis	Normalized values
Star Coordinate	Visual comparison	Radial display
Visual Method	Design problems	Solution visualization

Concept: All Pairwise Objective Plots

Scatter-Plot Matrix Method

For M Objectives:

- $\binom{M}{2}$ distinct plots ($M(M-1)$ if mirrored plots are counted)
- Each plot shows two objectives
- Diagonal shows objective ranges

Application:

- Upper triangle: one algorithm
- Lower triangle: another algorithm
- Allows direct comparison

	f_1	f_2	f_3
f_1	—	(f_1, f_2)	(f_1, f_3)
f_2	(f_2, f_1)	—	(f_2, f_3)
f_3	(f_3, f_1)	(f_3, f_2)	—

Note: the (i, j) plot is the mirrored version of the (j, i) plot.

Concept: Cross-Line Representation

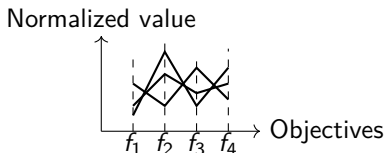
Value Path Method

Construction:

- 1 Horizontal axis: M objective identifiers ($1, 2, \dots, M$)
- 2 Vertical axis: normalized objective values
- 3 Each solution = line connecting objective values

Information Provided:

- **Spread:** how well solutions cover objective ranges
- **Trade-offs:** zig-zag patterns indicate conflicts
- **Diversity:** histogram can be superimposed



Two Distinct Goals

Performance Metrics for MOEAs

Goal 1: Convergence

- Solutions as close as possible to the Pareto-optimal front
- Measured by closeness metrics

Goal 2: Diversity

- Solutions as spread out as possible
- Measured by diversity metrics

Why Two Metrics Are Needed:

Algorithm	Convergence	Diversity
Algorithm A	✓ Good	× Poor
Algorithm B	× Poor	✓ Good

Neither dominates the other — both goals are essential!

Convergence Metrics

Metrics for Closeness to the Pareto-Optimal Front

1. Error Ratio (ER)

$$ER = \frac{1}{|Q|} \sum_{i=1}^{|Q|} e_i$$

- $e_i = 1$ if solution i is **not** Pareto-optimal; $e_i = 0$ otherwise
- **Lower is better** ($ER = 0 \Rightarrow$ all solutions are Pareto-optimal)

2. Generational Distance (GD)

$$GD = \frac{\left(\sum_{i=1}^{|Q|} d_i^p \right)^{1/p}}{|Q|}$$

- d_i : Euclidean distance from solution i to its nearest Pareto-optimal solution
- Usually $p = 2$ is used
- **Lower is better**

Diversity Metrics

Metrics for Diversity Among Solutions

1. Spacing (S)

$$S = \sqrt{\frac{1}{|Q| - 1} \sum_{i=1}^{|Q|} (d_i - \bar{d})^2}$$

- d_i : distance to nearest neighbor; $\bar{d}$: mean of all d_i
- **Smaller $\Rightarrow$ more uniform spacing**

2. Spread (Δ)

$$\Delta = \frac{\sum_{m=1}^M d_m^e + \sum_{i=1}^{|Q|-1} |d_i - \bar{d}|}{\sum_{m=1}^M d_m^e + (|Q| - 1) \bar{d}}$$

- d_m^e : distance to the extreme (boundary) Pareto-optimal solutions
- d_i : consecutive distance between neighboring solutions
- $\Delta = 0$: ideal uniform distribution covering all extremes

Combined Metrics

Metrics for Both Closeness and Diversity

1. Hypervolume (HV)

$$\text{HV} = \text{volume} \left(\bigcup_{i=1}^{|Q|} v_i \right)$$

- v_i : hypercube defined by reference point W and solution i
- **Larger is better**

2. Attainment Surface — Statistical Comparison

- 1 Choose diagonal cross-lines
- 2 For each algorithm, find intersection points with the front
- 3 Build frequency distribution of intersection points
- 4 Apply statistical test (e.g., Mann–Whitney)
- 5 Determine which algorithm is better

Output: $[a, b]$ where $a = \%$ where Algorithm A is better, $b = \%$ where Algorithm B is better, and $100 - (a + b) = \%$ inconclusive.

Why Systematic Test Problems?

Designing Test Problems for MOEAs

Problem with Existing Tests:

- Difficulty cannot be controlled
- Pareto-optimal front often unknown
- Cannot test specific algorithm features

The Three-Function Construction:

$$f_1(x) = f_1(x_1, \dots, x_k), \quad f_2(x) = g(x_{k+1}, \dots, x_n) \cdot h(f_1, g)$$

Role of Each Function:

Function	Controls	Tests
f_1	Diversity along front	Finding spread
g	Convergence difficulty	Reaching Pareto front
h	Pareto front shape	Handling convexity

Six Problems — Increasing Difficulty

Zitzler-Deb-Thiele (ZDT) Test Problems

Problem	Variables	Features
ZDT1	30	Convex Pareto front
ZDT2	30	Nonconvex Pareto front
ZDT3	30	Disconnected Pareto fronts
ZDT4	10	Many local fronts (21^9 of them!)
ZDT5	11	Deceptive (Boolean)
ZDT6	10	Non-uniform density + nonconvex

Key Insight:

- ZDT4 has $21^9 \approx 8 \times 10^{11}$ local Pareto-optimal fronts
- Only **one** of them is the global front
- Tests the algorithm's ability to avoid local optima

Difficulties in Constrained MOEAs

Constrained Test Problems

1. Difficulty Near the Pareto-Optimal Front

- Constraints eliminate parts of the unconstrained Pareto front
- Solutions must lie on constraint boundaries
- Requires correlation among decision variables

2. Difficulty in the Entire Search Space

- Infeasible bands/regions
- Algorithm must navigate through infeasible space
- Widening infeasible regions increases difficulty

The CTP Generator:

$$\cos(\theta)(f_2(x) - e) - \sin(\theta)f_1(x) \geq a \left| \sin \left\{ b\pi [\sin(\theta)(f_2(x) - e) + \cos(\theta)f_1(x)]^c \right\} \right|^d$$

Parameters: θ (rotation angle), a, b, c, d (shape of infeasible regions), e (translation).

Zitzler-Deb-Thiele Study (2000)

Comparative Studies of MOEAs

Algorithms Compared:

- 1 RAND — random search
- 2 SOGA — single-objective GA (weighted sum)
- 3 VEGA — Vector Evaluated GA
- 4 HLGA — weight-based GA
- 5 MOGA — Multi-Objective GA
- 6 **NSGA** — Non-dominated Sorting GA
- 7 NPGA — Niche Pareto GA
- 8 **SPEA** — Strength Pareto EA (only elitist)

Key Findings:

- 1 **All MOEAs outperform RAND**
- 2 **NSGA outperforms all non-elitist algorithms**
- 3 **SPEA (elitist) outperforms all algorithms**
- 4 Hierarchy: $\text{SPEA} > \text{NSGA} > \text{VEGA} > \text{NPGA}, \text{HLGA} > \text{MOGA} > \text{RAND}$
- 5 **Elitism is crucial** for good performance

Deb-Agrawal-Pratap-Meyarivan Study (2000)

Comparative Studies (Continued)

Algorithms Compared:

- NSGA-II (real-parameter)
- NSGA-II (binary-coded)
- SPEA
- PAES

Results Summary:

Metric	Best Algorithm
Convergence (GD)	NSGA-II (real) on most problems
Diversity (Spread Δ)	NSGA-II (real) on all problems
Overall	NSGA-II (real) consistently best

Key Insight: real-parameter NSGA-II with the SBX operator outperforms binary-coded versions, SPEA and PAES — especially on problems with nonconvexity, discontinuity, and multi-modality.

Which Space Should We Niche In?

Objective vs Decision-Space Niching

Objective-Space Niching:

- Focus: diversity in objective function values
- Advantage: works with any representation
- Disadvantage: may miss diverse decision variables

Decision-Space Niching:

- Focus: diversity in decision variable values
- Advantage: better for understanding solutions
- Disadvantage: representation-dependent

Example Problem:

$$f_1(x) = 1 - \exp(-4x_1) \sin^4(5\pi x_1), \quad f_2(x) = g(x_2) \left[1 - \left(\frac{f_1(x)}{g(x_2)} \right)^4 \right]$$

Observation: decision-space niching finds diverse x_1 values, while objective-space niching finds only $\sim 50\%$ of the x_1 space — **the choice depends on problem requirements.**

Post-Optimal Techniques

Searching for Preferred Solutions

1. Compromise Programming

- Find the solution closest to a reference point (ideal solution)
- Distance metrics: L_p -metric or Tchebycheff

2. Marginal Rate of Substitution

- Improvement in one objective per unit sacrifice in another
- Choose the solution with maximum rate

3. Pseudo-Weight Vector Approach

$$w_i = \frac{(f_i^{\max} - f_i(x)) / (f_i^{\max} - f_i^{\min})}{\sum_{j=1}^M (f_j^{\max} - f_j(x)) / (f_j^{\max} - f_j^{\min})}$$

- Relative importance of each objective for a solution
- Choose the solution closest to the user-preferred weight vector

Optimization-Level Techniques

Searching for Preferred Solutions

1. Biased Sharing Approach

$$d'_{ij} = \sqrt{\sum_{k=1}^M w'_k \left(\frac{f_k^{(i)} - f_k^{(j)}}{f_k^{\max} - f_k^{\min}} \right)^2}$$

- Higher weight $w'_k \Rightarrow$ more solutions near that objective's optimum

2. Guided Domination Approach

$$D_i(f(x)) = f_i(x) + \sum_{j \neq i} a_{ij} f_j(x)$$

- Controls which Pareto region is found
- Larger $a_{ij} \Rightarrow$ more emphasis on f_i

3. Weighted Domination Approach

$$\sum_{i=1}^M w_i l_i(x^{(1)}, x^{(2)}) \geq 1$$

- Generalizes domination with user preferences

Constrained Single-Objective Optimization

Exploiting MOEAs for Constrained Optimization

Reformulation as Multi-Objective:

$$\text{Minimize } f(x), \quad \text{Minimize } (g_j(x))^2, \quad j = 1, \dots, J$$

Advantages:

- 1 No penalty parameters needed
- 2 Natural handling of “soft” constraints
- 3 Provides trade-off between objective and constraint violations

Implementation: COMOGA

- Two objectives: original fitness + Pareto-ranking of constraint violations
- Parameter P_{cost} controls emphasis
- Dynamically adjusted to maintain feasible proportion

Goal Programming Reformulation

Exploiting MOEAs for Goal Programming

Conversion to Multi-Objective:

Goal Type	Original	Reformulated Objective
$\leq$	$f_i(x) \leq t_i$	Minimize $(f_i(x) - t_i)$
$\geq$	$f_i(x) \geq t_i$	Minimize $(t_i - f_i(x))$
$=$	$f_i(x) = t_i$	Minimize $ f_i(x) - t_i $

Advantages:

- 1 **No weight vector needed** upfront
- 2 **Multiple solutions** found simultaneously
- 3 Solutions correspond to different weight factors
- 4 Pseudo-weight vectors help choose the final solution

Application Example: welded beam design with cost and deflection goals — found multiple trade-off solutions in a single run.

Challenge: Increasing Number of Objectives

Scaling Issues in MOEAs

As M increases, more solutions become non-dominated:

Population Size $N = 100$	% Non-Dominated
$M = 2$	$\sim 10\%$
$M = 5$	$\sim 40\%$
$M = 10$	$\sim 80\%$

Implications:

- 1 **Less selection pressure** (all solutions have similar fitness)
- 2 **Elitism becomes problematic** (too many elites)
- 3 **Need larger population size** to maintain diversity

Minimum Population Size ($\leq 30\%$ non-dominated):

$M = 2$: $N \approx 20$; $M = 5$: $N \approx 100$; $M = 10$: $N \approx 700$

Population size grows rapidly with M !

Theoretical Foundations

Convergence Issues in MOEAs

Rudolph's Convergence Work — Key Condition: Positive Variation Kernel

- Probability of creating any solution from any parent set is > 0
- Guarantees eventual discovery of Pareto-optimal solutions

Base Algorithm (Infinite Archive):

- 1 Generate population from the elite set
- 2 Combine population + elite set
- 3 New elite = non-dominated solutions of the combined set
- 4 **Property:** all Pareto-optimal solutions eventually found

Problem with Finite Archive:

- Once the archive is full with N Pareto-optimal solutions, no new solutions are accepted
- **No guarantee of diversity!**

Open Challenge: design an MOEA with both a convergence proof *and* a diversity guarantee.

Problem: Too Much Elitism

Controlled Elitism in NSGA-II

Original NSGA-II:

- Accepts all best non-dominated solutions
- May lose **lateral diversity**
- Can get stuck on local Pareto-optimal fronts

Solution: Geometric Distribution

$$N_i = N \cdot \frac{1-r}{1-r^K} \cdot r^{i-1}$$

- N_i : max individuals allowed in front i ; r : reduction rate ($r < 1$); K : number of fronts

Effect: smaller $r \Rightarrow$ more exploration (more fronts represented); larger $r \Rightarrow$ more exploitation (fewer fronts). The optimal r balances both.

Results: controlled NSGA-II found $r = 0.65$ best for the test problem, maintaining 20–25 solutions in front 1 (vs. 100 in the original) with better convergence to the true Pareto-optimal front.

1 Representation

- Multiple methods for visualizing > 2 objectives; scatter-plot matrix most common

2 Performance Metrics

- **Need at least two metrics:** convergence + diversity; no single metric evaluates both goals

3 Test Problems

- Systematic design essential for fair comparison; ZDT problems are widely used benchmarks

4 Comparative Studies

- **Elitism is crucial** for good performance; NSGA-II (real-parameter) consistently strong

5 Key Research Issues

- Scaling to many objectives; convergence with diversity guarantee
- Controlled elitism for lateral diversity; finding preferred solutions

Key References

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